Code No: 34022 **R05** **Set No. 2**

**II B.Tech II Semester Supplimentary Examinations,February 2010**
**SOFTWARE ENGINEERING**
**Common to Information Technology, Computer Science And Engineering, Computer Science And Systems Engineering**

**Time: 3 hours** **Max Marks: 80**

**Answer any FIVE Questions**
**All Questions carry equal marks**

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1. (a) How would you assess design quality in the formal technical review?
   (b) Elaborate on FURPS quality attributes. [8+8]
2. Discuss about requirements elicitation and analysis process in detail. [16]
3. (a) What is the overall strategy for software testing? Explain it clearly. [10]
   (b) Discuss a testing strategy for Object-Oriented architectures. [6]
4. (a) Discuss the seven principles of risk management which were identified by SEI. [10]
   (b) Distinguish between generic risks and product specific risks. [6]
5. (a) Discuss in detail about software safety. [6]
   (b) Discuss the ISO 9001:2000 standard. [10]
6. (a) Discuss about design evolution . [8]
   (b) Draw and explain the state diagram for weather station. [8]
7. (a) Discuss the merits and demerits of spiral model. Differentiate it with other models.
   (b) Elaborate the conditions under which spiral model can be applied. [16]
8. Is there ever a case when the generic activities of the software engineering process don't apply? If so, describe it. [16]

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Code No: 34022 **R05** **Set No. 4**

**II B.Tech II Semester Supplimentary Examinations,February 2010**
**SOFTWARE ENGINEERING**
**Common to Information Technology, Computer Science And Engineering, Computer Science And Systems Engineering**

**Time: 3 hours** **Max Marks: 80**

**Answer any FIVE Questions**
**All Questions carry equal marks**

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1. Who should be involved in a requirements review? Draw a process model showing how a requirements review might be organized. [6+10]

2. Discuss in detail about Formal Technical Reviews(FTR). [16]

3. List different types of requirements. Also explain the purpose of them. [8+8]

4. (a) What is software architecture? Discuss how it's components are derived.
   (b) Give an overall architectural structure for Safe Home system with top-level components. **[16]**

5. (a) Describe the benefits that smoke testing provide, when it is applied on complex, time critical software engineering projects. [6]
   (b) What is a "Critical Module" and why should we identify it? [6]
   (c) Discuss about integration test documentation. [4]

6. Discuss various process maturity levels. Also discuss various KPAs that must be achieved in each level. [16]

7. Describe the best and worst interfaces that you have ever worked with and critique them relative to the concepts of user interface design. [16]

8. (a) Describe five software application areas in which software safety and hazard analysis would be a major concern. [10]
   (b) Can you think of a situation in which a high probability, high-impact risk could not be considered as a part of your RMMM plan? [6]

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Code No: 34022 **R05** **Set No. 1**

**II B.Tech II Semester Supplimentary Examinations,February 2010**
**SOFTWARE ENGINEERING**
**Common to Information Technology, Computer Science And Engineering, Computer Science And Systems Engineering**

**Time: 3 hours** **Max Marks: 80**

**Answer any FIVE Questions**
**All Questions carry equal marks**

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1. (a) What is data flow model? Explain the notations used in it.
   (b) Discuss about the semantic data model of a software design. [3+3+10]
2. (a) Define user and system requirements. Differentiate between them.
   (b) What is requirements specification? Who are the readers of different types of requirements specification? [4+2+3+7]
3. Discuss clearly about risk projection. [16]
4. (a) What is meant by use case description? Explain with an example. [6]
   (b) What is Architectural design? State and explain the three layers in the weather station software. [10]
5. (a) What is team software process ? Discuss it's objectives.
   (b) Define the frame work activities of TSP. [3+3+10]
6. (a) Illustrate about component level design elements in detail.
   (b) What is deployment diagram? Draw a deployment diagram for Safe Home system in UML. [6+4+6]
7. (a) Why is it that a single, all-encompassing metric can not be developed for program complexity or program quality? [8]
   (b) Software for a system X has 24 individual functional requirements and 14 nonfunctional requirements. What is the specificity of the requirements the completeness? [8]
8. A Formal Technical Review (FTR) effective only if every one has prepared in advance. How do you recognize a review participant who has not prepared? What do you do if you are the review leader? [16]

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Code No: 34022 **R05** **Set No. 3**

**II B.Tech II Semester Supplimentary Examinations,February 2010**
**SOFTWARE ENGINEERING**
**Common to Information Technology, Computer Science And Engineering, Computer Science And Systems Engineering**

**Time: 3 hours** **Max Marks: 80**

**Answer any FIVE Questions**
**All Questions carry equal marks**

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1. What is meant by risk projection? Discuss in detail the various steps in risk projection. [16]

2. (a) What is data design. Explain how is it done with examples.
   (b) Elaborate on architectural design elements. [2+6+8]

3. (a) Define brain storming. Explain where it is used with an example.
   (b) What are view point and service template forms? Explain why they are used? [6+10]

4. Recommend the life cycle model( waterfall or evolutionary ) for the following software systems.
   a) Payroll.
   b) Business accounting.
   c) Point-of-sale software. [5+5+6]

5. (a) Discuss about ISO 9000 quality standards. [8]
   (b) Discuss about software quality management tools. [8]

6. (a) The software analysis and design are constructive tasks, and software testing is considered to be destructive from the point of view of developer. Discuss. [8]
   (b) Who will test the software, either developer or an independent test group? Discuss the advantage and draw backs of each one. [8]

7. (a) State some examples that illustrates why response time variability of user interface can be an issue. [8]
   (b) Discuss the design principles that reduce the user?s memory load in user interface. [8]

8. Suggest three specialized software tools which might be developed to support a Process improvement programme in an organization. [16]

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